

Fluid Bed Coating: A Digital Twin

PHYSICS-BASED PROCESS MODEL WITH DOE-FITTED CORRELATIONS

FROM MECHANISTIC ODE SYSTEM TO A DEPLOYED NO-CODE TOOL

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Context & Problem

Fluid-bed coating controls **drug release** (dissolution), a **critical quality attribute** (cQA)

Observed **lack of process performance** → **low capability** to meet dissolution specifications

Process optimization today is **experimental, slow** and **costly**: material, equipment time and testing for every trial run

Many **coupled parameters** → a large design space that is impractical to explore by trial-and-error or DoE at large scale (estimated effort: 60 full-scale runs → material cost €1.2M)

Idea: a digital twin that predicts coating weight gain → dissolution from the process settings

Value: screen process changes virtually & run **only the experiments that matter**

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Process and Variables

Three sequential stages: pre-heating → spraying → drying

- each stage's end-state seeds the next

Inputs / controls (blue):

- Inlet air flow & temperature
- Spray rate, coating concentration, solution quantity, atomization air pressure
- Batch size

Disturbances / noise (purple):

- Inlet air humidity, material variability (SSA, diameter)

State variables (green):

- Product temperature, particle solvent fraction, coating mass
- Gas solvent fraction, gas temperature

Output (CQA):

- Coating weight gain → **Dissolution profile**

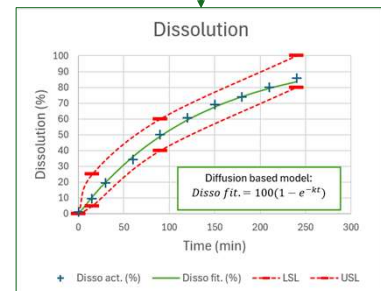
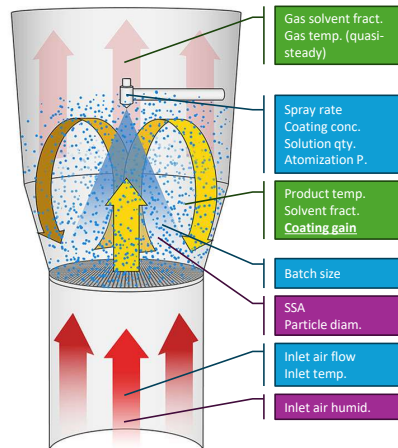


Illustration from <https://foodfeedfinechemicals.glatt.com/technology-know-how/technologies/fluid-bed-technology/>

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Modeling strategy: ODE system & Dissolution model

Each stage = coupled ODEs from **energy & mass balances**; the stages are solved in sequence

Energy-balance ODE: how does heat transfer between phases?

- Heat transfers from hot air to the particles and to the spray-droplet solvent
- Drives solvent evaporation that dries the coating film on the particle surface

Mass-balance ODE: how do solvent and coating evolve?

- Tracks solvent concentration in the air and at the granule surface
- Tracks deposited film-coating

Closure physics:

- Heat/mass-transfer correlations
- Drying rate from vapor pressure + sorption isotherm

Dissolution model:

- Weight gain
- first-order release rate constant k – **inversely proportional to coating weight gain**
- predicted dissolution curve

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Demo: Physics skeleton (notebook 05a)

Full pre-heat → spray → dry chain

Produces temperature & moisture trajectories, final weight gain and the dissolution curve

The catch:

2 coating-loss terms r_{spraying} and r_{drying} are still **free parameters**

→ model is **descriptive, not predictive**



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Challenge: 2 unknown loss parameters

The ODEs need two empirical parameters:

- r_{spraying} → spray-drying loss
- r_{drying} → attrition loss

These are **not fundamental constants**

They depend on **process settings, material and environment**

Gap: we need correlations $r = f(\text{process inputs})$ so the digital twin runs on settings alone
Otherwise, they must be re-fitted for every run, which defeats a predictive twin.

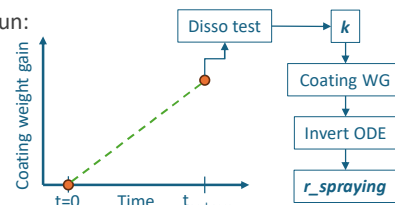
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Solution: Small-scale DoE + inverse modeling

Inverse-modeling pipeline recovers the true loss parameters per run:

- fit dissolution model $\rightarrow k$
- $\rightarrow$ invert dissolution: estimate weight gain from k
- $\rightarrow$ invert spraying and drying ODE systems
- $\rightarrow$ recover r_{spraying} and r_{drying} for each DoE run



Structured Design of Experiments (20 runs) spans the input space efficiently at small scale

Regress the recovered r_{spraying} / r_{drying} on the design space process factors $\rightarrow$ **empirical correlations:**

$$r_{\text{spraying}} = f(\text{spray rate, coating concentration, dry-matter ratio})$$

$$r_{\text{drying}} = f(\text{batch size, dry-matter ratio, SSA, air humidity})$$

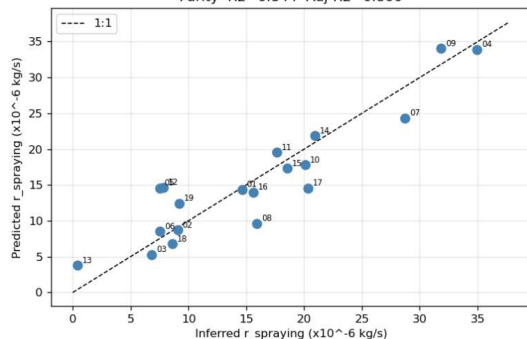
Scope: valid only inside the DoE design space (extrapolation is unvalidated)

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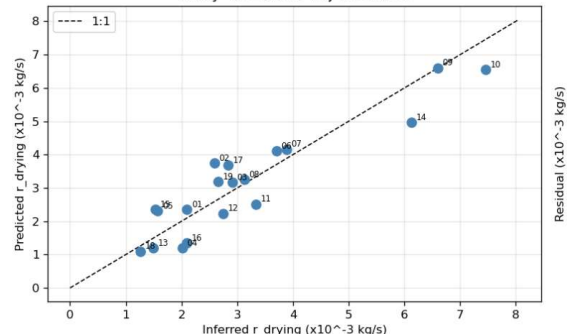
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r_{spraying} and r_{drying} : Actual vs. Correlation Fit

Step 6a -- r_{spraying} selected model
 Spray rate + Coating conc + DM ratio + Coating conc x DM ratio
 Parity R2=0.844 Adj-R2=0.800



Step 6b -- r_{drying} selected model
 Main: Batch size (kg), DM ratio (g/kg), SSA (cm2/g), Inlet humidity (g/kg)
 Interaction: DM ratio x Inlet humidity
 Parity R2=0.851 Adj-R2=0.794



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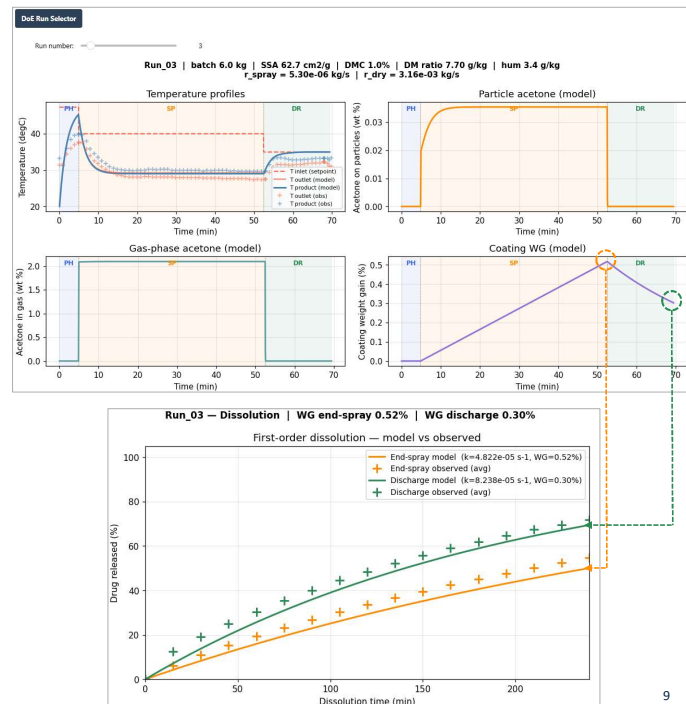
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Validation against DoE runs (notebook 06)

Twin run across all DoE runs using only the process settings

Predicted vs. measured: temperature and dissolution, run by run

Establishes confidence within the design space



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The complete digital twin (notebook 05b)

Same physics as 05a, but r_{spraying} and r_{drying} come from correlations

Fully **predictive** from process settings, no manual tuning

Enter settings → trajectories, final weight gain and dissolution curve, end-to-end

Humidity entered in real units (g/kg) as a studied disturbance



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Deployment: No-code tool for concrete value

The notebook digital twin turned into an **interactive web app**

No Python and no setup for the end user, just a link:
<https://huggingface.co/spaces/SamdGuizani/fluid-bed-coating-twin>

Screen “what-if” process changes virtually before any lab trial

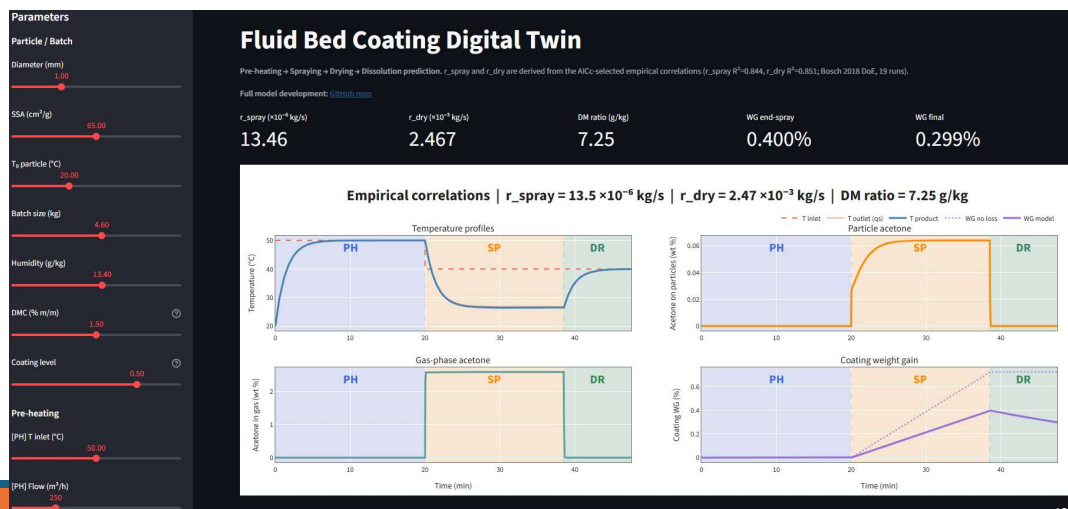
Value: fewer physical experiments → lower cost, faster development, less material / API consumed



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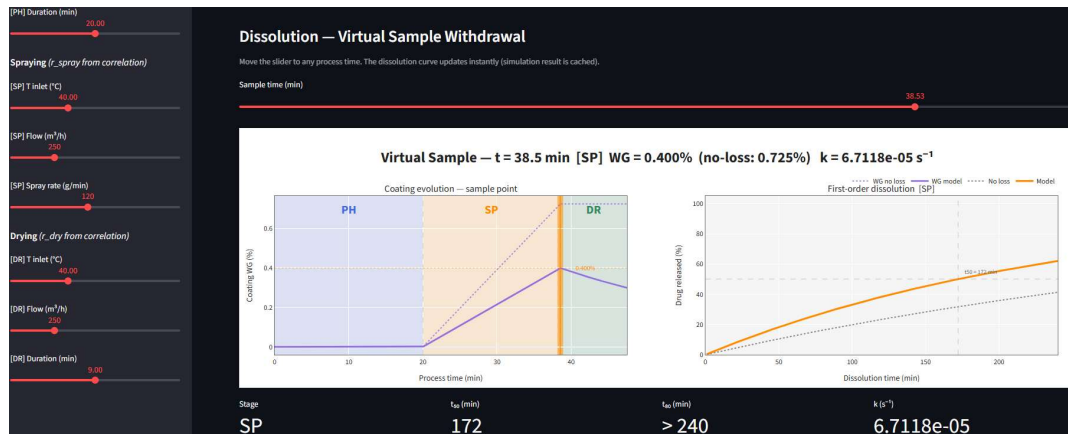
Web app screenshot (1)



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Web app screenshot (2)



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Further applications & Vision

Scale-up: rebuild *r_spraying* and *r_drying* correlations at large scale, limiting the experiments to:

- relevant process inputs for the large scale
- only inputs that were found influential at small scale

Grid search / optimization: auto-sweep the design space for the settings that hit a target dissolution profile

Sensitivity analysis: rank the most influential inputs → focus experiments and tighten control on the critical ones

Adaptive control: use the observed in-process state to re-tune the parameters toward the target quality (soft sensor / closed loop)

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Key takeaways

Built a **physics-based digital twin** of fluid-bed coating (pre-heat → spray → dry → dissolution)

Closed the predictive gap with DoE-fitted correlations for the two loss parameters → **Hybrid model**

Validated within the design space and **deployed as a no-code app**

Enables **virtual process optimization** → less cost, less experimentation

Resources:

- https://github.com/SamdGuizani/FB_Coating_Digital_Twin/tree/main
- <https://huggingface.co/spaces/SamdGuizani/fluid-bed-coating-twin>

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Thank you!

Time for Questions...

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Backup Slides

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Pre-heating stage & shared sub-models

State variable [T_p]: convective heating of the dry bed

Energy: $\frac{dT_p}{dt} = \frac{\alpha_h A_{tot} (T_g - T_p)}{M_b c_{p,p}}$

Solver: RK45 (non-stiff) · IC: $T_p = \text{ambient}$ (~293 K) → end-state seeds spraying

Transfer coefficients: Ranz-Marshall (shared by all stages)

$$u = \frac{\dot{m}_{air}}{\rho_{air} A_{bed}} \text{Re} = \frac{\rho_{air} u d_{eq}}{\mu_{air}} \text{Pr} = \frac{c_{p,air} \mu_{air}}{\lambda_{air}} \text{Sc} = \frac{\mu_{air}}{\rho_{air} D_{ac}}$$

$$\text{Nu} = 2 + \text{Pr}^{2/5} (0.43 \text{Re}^{1/2} + 0.06 \text{Re}^{2/3})$$

$$\text{Sh} = 2 + \text{Sc}^{2/5} (0.43 \text{Re}^{1/2} + 0.06 \text{Re}^{2/3})$$

$$\alpha_h = \frac{\text{Nu} \lambda_{air}}{d_{eq}} \quad \alpha_m = \frac{\text{Sh} D_{ac}}{d_{eq}} \quad A_{tot} = N \pi d_{eq}^2 \quad \text{or} \quad \text{SSA} \cdot M_b$$

α_h [W m⁻² K⁻¹] · α_m [m s⁻¹] · $N = M_b / m_p$ · spraying: add atomisation air to $\dot{m}_{air}$

Quasi-steady gas temperature: Effectiveness – Number of Transfer Units (NTU)

$$\text{NTU} = \frac{\alpha_h A_{tot}}{\dot{m}_{air} c_{p,air}} ; T_g = \frac{T_{g,in} + \text{NTU} \cdot T_p}{1 + \text{NTU}}$$

Coherent with MODELLING_BACKGROUND_README.md §3.1–3.3, §4.1

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Spraying stage

State variables [Y_p, Y_g, M_c, T_p] (BDF solve, stiff)

Spray split dry matter: $\dot{m}_{DM} = \dot{m}_{spray} x_{DM}$

Spray split solvent: $\dot{m}_{solv} = \dot{m}_{spray}(1 - x_{DM})$

ODE system

$$\text{Particle solvent: } \frac{dY_p}{dt} = \frac{\dot{m}_{solv}}{M_b} - R_D$$

$$\text{Gas solvent: } \frac{dY_g}{dt} = \frac{R_D M_b - \dot{m}_{air}(Y_g - Y_{g,in})}{\dot{m}_{air} \tau_g} \quad (\tau_g = 0.5s)$$

$$\text{Coating mass: } \frac{dM_c}{dt} = \dot{m}_{spray} x_{DM} - r_{spraying} \text{ (order-0 loss = spray-drying)}$$

$$\text{Energy: } \frac{dT_p}{dt} = \frac{\alpha_h A_{tot}(T_g - T_p) - R_D M_b \Delta h_{vap}}{M_b(c_{p,p} + Y_p c_{p,ac}^{liq})}$$

R_D = specific evaporation rate [kg acetone · kg⁻¹ · s⁻¹] — defined on the Drying slide (§3.4)

Empirical loss correlation $r_{spraying}$ (Step 06a, $R^2 = 0.84$)

$$10^6 r_{spraying} [\text{kg s}^{-1}] = 15.334 + 0.197SR^* + 9.204CC^* - 2.799DM^* - 3.698(CC^* \cdot DM^*)$$

centred predictor $x^ = x - \bar{x}_{train}$ · SR spray rate, CC coating conc., DM dry-matter ratio · clipped ≥ 0

Coherent with MODELLING_BACKGROUND_README.md §4.2, §6.6

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Drying stage

State variables [Y_p, Y_g, M_c, T_p] (spray off, BDF solver)

$$\text{Particle solvent: } \frac{dY_p}{dt} = -R_D$$

$$\text{Gas solvent: } \frac{dY_g}{dt} = \frac{R_D M_b - \dot{m}_{air}(Y_g - Y_{g,in})}{\dot{m}_{air} \tau_g}$$

$$\text{Coating attrition: } \frac{dM_c}{dt} = -r_{drying} \left(\frac{M_c}{M_b} \right)^n \quad (n = 1)$$

Energy balance: identical in form to the spraying stage

Drying rate R_D (evaporation sub-model, §3.4)

$$\log_{10} P_{ac}^{sat}(T_p) = A - \frac{B}{C + T_p} \quad [^\circ\text{C}] \rightarrow \text{Pa} \times 133.322 (A = 7.1327, B = 1219.97, C = 230.653)$$

$$p_{ac}(Y_g) = P_{tot} \frac{Y_g}{MW_{ac} + Y_g}; \quad \varphi(Y_p) = \frac{Y_p}{|Y_p| + \alpha_L} \quad (\alpha_L = 0.05)$$

$$R_D = \varphi \frac{\alpha_m A_p [P_{ac}^{sat} - P_{ac}]^+}{m_p (R/MW_{ac}) \bar{T}}; \quad \bar{T} = \frac{1}{2} (T_p + T_g)$$

Empirical loss correlation r_{drying} (Step 06b · $R^2 = 0.85$)

$$10^3 r_{drying} [\text{kg s}^{-1}] = 3.102 + 0.8133BS^* - 0.7280DM^* + 0.1835SSA^* - 0.0720H^* + 0.0339(DM^* \cdot H^*)$$

centred $x^ = x - \bar{x}_{train}$ · BS batch size, SSA surface area, H inlet humidity · humidity enters the twin only here

Coherent with MODELLING_BACKGROUND_README.md §3.4, §4.3, §6.6

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Dissolution model

Retained model: first-order release

$$F(t) = 100(1 - e^{-kt}) \quad [\% \text{ released}]$$

Mechanistic link to the coating: diffusion through the coating film

$$k = \frac{m_{\text{sample}} \text{SSA}^2 P \rho_c}{V_{\text{disso}} x_c} \quad [\text{s}^{-1}] \quad x_c = \frac{M_c}{M_b} \quad (\text{Coating mass fraction})$$

Constants: $m_{\text{sample}} = 1.058 \text{ g}$ · $P = 1.5 \times 10^{-7} \text{ cm}^2 \text{ s}^{-1}$ (film permeability) · $\rho_c = 0.4 \text{ g cm}^{-3}$ · $V_{\text{disso}} = 1000 \text{ mL}$ · $\text{SSA in cm}^2 \text{ g}^{-1}$
 $\Rightarrow$ more coating $\rightarrow$ larger $x_c \rightarrow$ smaller $k \rightarrow$ slower release

Other candidate models (fitted for comparison; first-order retained)

Zero-order: $F = kt$ Higuchi: $F = k\sqrt{t}$ Korsmeyer-Peppas: $F = 100kt^n$

Coherent with MODELLING_BACKGROUND_README.md §5